

CLASS 8 · ICSE · SELINA CONCISE MATHEMATICS · DPP SHEET

Chapter 2: Exponents (Powers)

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. Evaluate $(-5)^3$ and $\left(\frac{1}{2}\right)^{-4}$.
2. Simplify $3^7 \div 3^{10} \times 3^5$.
3. Simplify $\left(\frac{4}{7}\right)^{-2} \times \left(\frac{4}{7}\right)^4$.
4. Evaluate $(125)^{\frac{2}{3}}$.
5. Evaluate $\left(\frac{16}{81}\right)^{\frac{3}{4}}$.
6. Simplify $(3^{-1} \times 4^{-1})^{-1} \times 5^{-1}$.
7. Find x if $2^{x-3} = 32$.
8. Find x if $5^{2x} \div 5^3 = 5^5$.
9. Write 0.0000073 in standard form.
10. Simplify $\left(\frac{1}{4}\right)^{-2} + \left(\frac{1}{2}\right)^{-2} + \left(\frac{1}{3}\right)^{-2}$.

Answer Key

1. -125 and 16

2. $3^2 = 9$

3. $\frac{16}{49}$

4. 25

5. $\frac{8}{27}$

6. $\frac{12}{5}$

7. $x = 8$

8. $x = 4$

9. 7.3×10^{-6}

10. $16 + 4 + 9 = 29$